

Premchand Goliya, chairman and managing director, Meco Instruments Pvt Ltd.

Coined for connectivity

Technology has never been so exciting before and it will get even more so as consumers continue to demand better ideas and innovation to help improve people's lives through end-to-end connectivity. "We are always on the lookout for devices that will feed our insatiable appetite for new gadgets such as smartphones, mini drones, cameras and other cool and trendy items that will make us feel good. What many people do not understand are the processes that each product has to go through to ensure usability and functionality for the intended purpose, and behind each of these steps—design, manufacture and test," says Tangri.

The Internet of Things (IoT) is the new buzzword coined for smart and connected electronic devices to enable greater efficiency and productivity in our daily lives. These devices aim to permeate into our homes, vehicles and buildings, managing security, safety, energy, inventories and even our well-being. Proliferation of smartdevices has begun to influence the design and use of T&M equipment.

Test techniques and solutions will continue to evolve to ensure that smartdevice developers and manufacturers can reduce time to market and ramp quickly to high-volume manufacturing, ensuring that all smartdevice consumers can get the high-quality product they want, when they want it.

"Connected lifestyles, machine-to-machine (M2M), smartcities and the IoT are some of the trends that will lead the growth of the T&M industry. Connected cars, connected lifestyles, higher data and lower cost of smartphones are some trends seen in the telecom market. Operators are using new technologies like MIMO, Vo LTE, LTE-Advanced and

STRENGTHS

- Continuous technological advancements in testing equipment like use of multi-functional testing kits
- Ongoing practice of procuring most-reliable testing equipment
- Adoption of best testing practices by OEMs
- Stringent quality and safety standards for end products

WEAKNESSES

- Enforcement of quality standards like harmonics level, etc not yet mandatory by the government
- Need to revive important quality norms like ISI

SWOT

OPPORTUNITIES

- Introduction of government initiatives like Make in India, Digital India and others
- Growth of domestic manufacturing in areas like defence, telecommunications and mobile
- Growth in power demand requires investment in reliable, accurate and efficient equipment
- Requirement of efficient generation, transmission and distribution of power
- Increasing awareness on predictive maintenance among manufacturers
- Proliferation of smartdevices

THREATS

- Lack of industry awareness about usability and functionality of testing equipment
- Various unscrupulous players not adhering to standard engineering and commercial practices
- Quality affected because government tender business done only on L1 basis
- Increasing focus on quality of end-use segments time consuming, which affects growth of this industry

SWOT analysis of T&M industry (Compiled by Sneha Ambastha, EYF Group)

LTE Carrier Aggregation to improve spectrum efficiency and customer experience of high data speed," says Tripathi. According to him, 4G roll out and 5G R&D in the telecom industry will create opportunities for the T&M industry.

"5G presents a number of new design and test challenges. While technologies that will constitute 5G are still being worked on, ITU-R has identified three main usage scenarios for 5G: enhanced mobile broadband with up to 20Gbps peak data rate, massive machine-type communications and ultra-reliable and low-latency communications.

"To meet the requirements for this next generation of mobile communications, researchers are looking to a wide range of enabling technologies in RF, microwave and millimeter-wave frequencies, ranging from advanced waveforms and multiple-access schemes to multiple-antenna techniques such as massive MIMO and beam-forming. Being the first and best in 5G depends on tools that

let you explore new signals, scenarios and topologies," explains Tangri.

Both Tangri and Tripathi indicate the unique opportunities related to the consumer electronics market in India. Consumer electronics have infiltrated every part of our lives and have had a broad influence in how we interact with everyday things. Whether it is telling your car's navigation system where you want to go or using a touchscreen to set up your refrigerator, new capabilities enable features that have not been previously possible in T&M equipment. Touchscreens have revolutionised the consumer electronics market and are working their way into the T&M environment.

"A combination of the right hardware with a design for touch UI enables us to use equipment easily and efficiently. Designs for touch UIs are crucial for creating an effective and usable touchscreen environment; it is not enough to simply add a touchscreen to an existing UI. The number-one goal of T&M equipment